

Datum
Date 29.01.2010
Date

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Sheet 1
Feuille

Anmelde-Nr:
Application No: 05 711 431.6
Demande n°:

The examination is being carried out on the **following application documents**

Description, Pages

1-66 as published

Claims, Numbers

1-42 filed with entry into the regional phase before the EPO

Drawings, Sheets

1-38 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure:

- D1 MATSUI ET AL: "Fatty acid hydroperoxide cleaving enzyme, hydroperoxide lyase, from tea leaves" PHYTOCHEMISTRY, PERGAMON PRESS, GB, vol. 30, no. 7, 1 January 1991 (1991-01-01) , pages 2109-2113
- D2 GENSLER HELEN L ET AL: "Prevention of photocarcinogenesis by topical administration of pure epigallocatechin gallate isolated from green tea" NUTRITION AND CANCER, vol. 26, no. 3, 1996 , pages 325-335
- D3 ZHAO J ET AL: "Photoprotective effect of black tea extracts against UVB-induced phototoxicity in skin." PHOTOCHEMISTRY AND PHOTOBIOLOGY OCT 1999, vol. 70, no. 4, October 1999 (1999-10) , pages 637-644
- D4 VAYALIL PRAVEEN K ET AL: "Treatment of green tea polyphenols in hydrophilic cream prevents UVB-induced oxidation of lipids and proteins, depletion of antioxidant enzymes and phosphorylation of MAPK proteins in SKH-1 hairless mouse skin." CARCINOGENESIS (OXFORD), vol. 24, no. 5, May 2003 (2003-05) , pages 927-936

The present application pertains to a method of fractionating green tea leaves (*Camellia sinensis*) and to use these fractions as photoprotective ingredient.

1 **Article 123(2) EP**

The present set of claims does not contravene Article 123(2) EPC.

2 **Clarity (Article 84 EPC) and Novelty (Article 54 EPC)**

The fractions claimed in claim 1-14 were obtained by the process illustrated in Figure 1. However, said claims are not formulated as product by process claims. Consequently, is any tea leaf membrane/ cytoplasmic fraction novelty destroying for the subject-matter of said claims.

D1 discloses on page 2110, 1. column the preparation of tea leave fractions. The leaves were homogenized with a buffer and centrifuged at 1.000 x g. The resultant supernatant was again centrifuged at 100.000 x g. The 1.000 x g pellet can be considered a cell wall fraction, the 1.000 x g supernatant cell juice, the 100.000 x g pellet a membrane fraction and the 100.000 x g supernatant cell juice serum plus cytoplasm. Considering that the buffer stabilises the pH, it can be seen as "stabilizer" (claim 15).

Consequently, is D1 novelty destroying for the subject-matter of claims 1-15.

3 **Inventive step (Article 56 EPC)**

3.1 The teaceae family comprises 40 genera and 600 species. The application discloses a skin protective effect only for the fraction of one single species, namely *Camellia sinensis*. It is unlikely that all other species have the same effect. Consequently, is the problem underlying the present application, the provision of a fraction capable of preventing photocarcinogenesis not fulfilled over the entire scope of the claimed subject-matter. The claims should be limited to *Camellia sinensis* and closely related plants.

3.2 The closest prior art is represented by the teaching of D4. The document reveals that green tea polyphenols (GTP) or its most chemopreventive constituent epigallocatechin-3-gallate (EGCG) protects skin from adverse effect of solar ultraviolet radiation when topically applied in a hydrophilic cream.

However, there is a difference between D4 using GTP or EGCG and the present application that uses chemically undefined cell fractions.

The technical problem to be solved by the present application therefore lies in the provision of alternative compounds for preventing photocarcinogenesis .

The problem has been solved by the provision of *Camellia sinensis* cell wall fraction, cell juice serum and membrane fraction.

For the assessment of inventive step, the question to be answered is whether it was obvious for a skilled person to have used fractions of tea leaves instead of purified GTP or EGCG.

GTP are well known antioxidants. It has been repeatedly demonstrated that oral and topical application of polyphenols before UVB irradiation protects mice skin from UV-induced skin cancer development (D3: sentence bridging page 637 and page 638).

Several polyphenols present in green tea have been shown to possess anticarcinogenic activity: the most active is EGCG, which is also the major constituent of GTP. D2 reveals in an animal model that the induction of skin tumours by UV radiation was significantly reduced by topical, but not by oral, administration of purified EGCG.

Treating the skin with green tea polyphenols has been shown to modulate biochemical pathways involved in inflammation, cellular proliferation, expression of tumour promoters, and UV light-induced inflammation in the skin.

The applicant detected the highest UV-protection property of Camellia products in the membrane fraction extract followed by cell juice serum and cell wall fraction extract which is identical to white tea extract (control).

The activity seen in white tea extract is most likely associated with the presence of polyphenols in this extract. It is therefore reasonable to assume that this activity which is inherent to the plant can be found once the plant is fractionated. Even black tea, produced by cutting Camellia leaves and allowing its fermentation can be used as source for polyphenols which, when topically administered, protect mice from UVB damage of skin (D3, page 642, last sentence).

In the present application no comparative experiments have been performed with GTP or purified EGCG that allow to conclude a superior effect of a Camellia cell fraction.

Fractionating green tea leaves and using these fractions for preventing photocarcinogenesis therefore seem not to involve inventive step in the sense of Article 56 EPC.

4 Third Party Observation under Article 115 EPC

A Third Party Observation has been filed with a letter dating 2.7.2009. The observation refers to novelty (Article 54 EPC) of the present application. The applicant is invited to consider said objection.

- 5** In view of the above objections, the applicant is asked to submit his comments and/ or a new set of claims.

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The attention of the applicant is drawn to the fact that the application may not be amended in such a way that it contains subject-matter which extends beyond the content of the application as filed (Article 123(3) EPC).

In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.

If the applicant regards it as appropriate these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed.